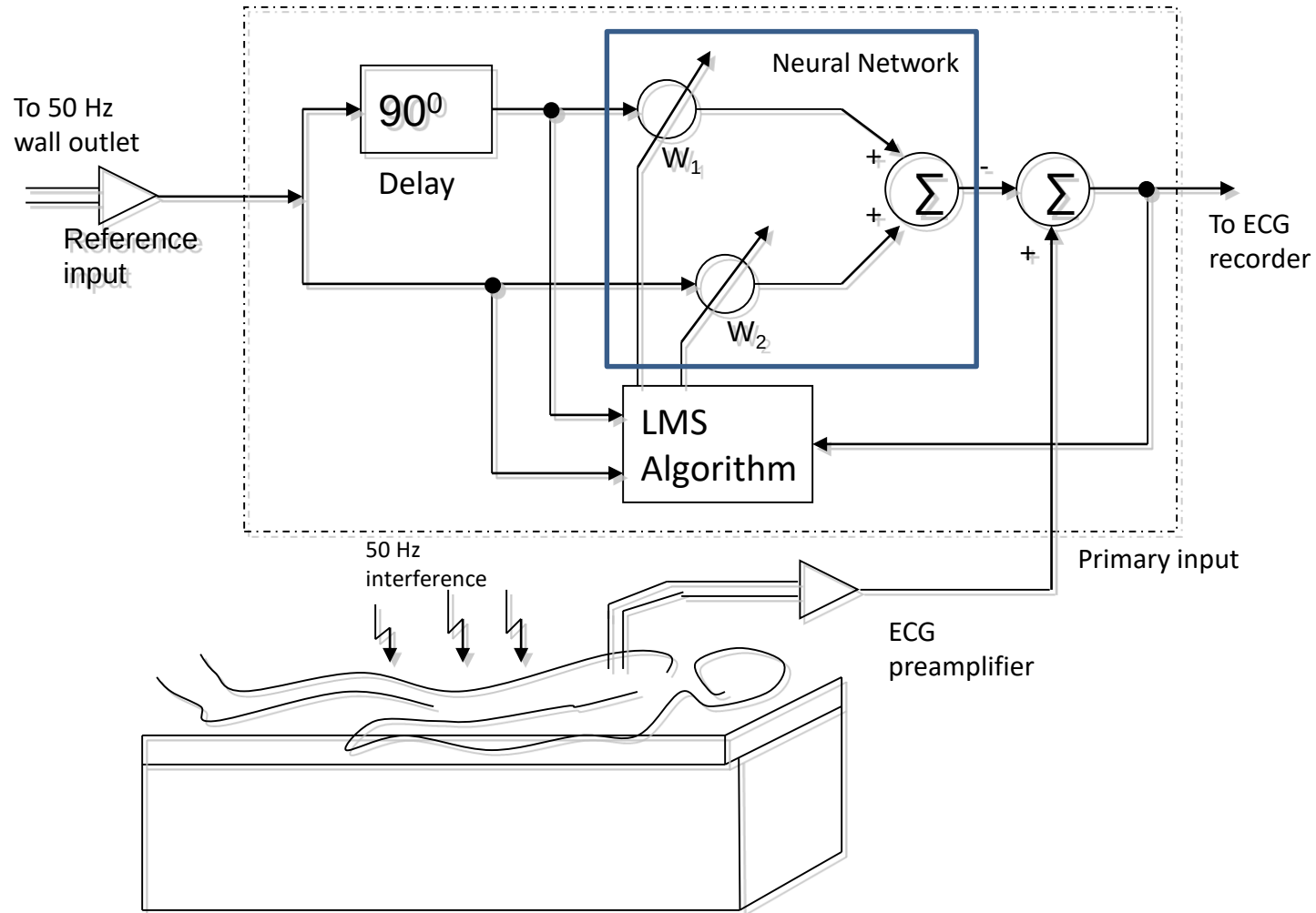


Applications of Neural Networks



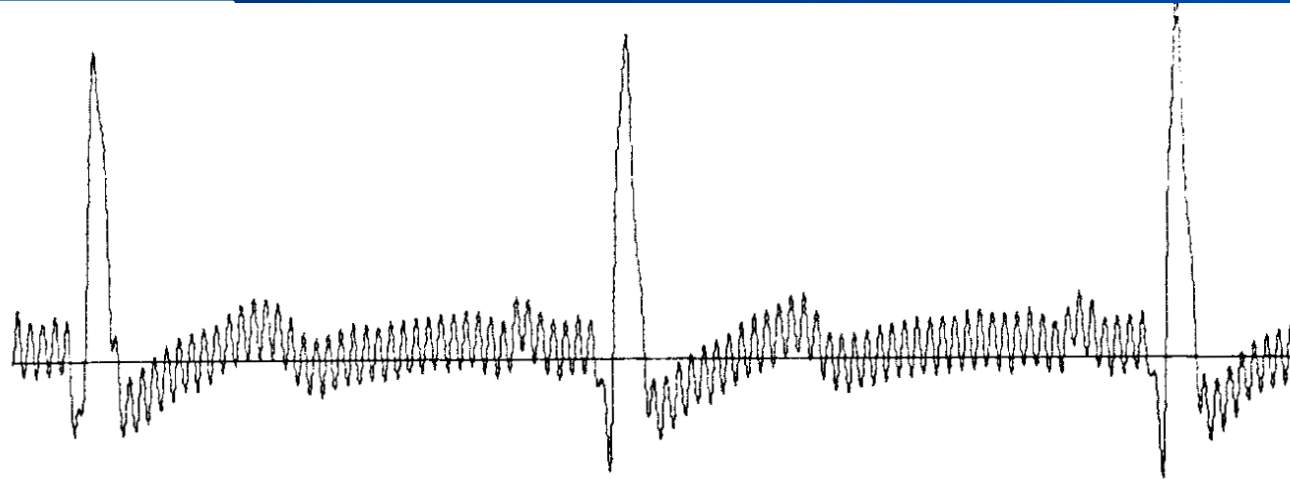
Dr. Satyasai Jagannath Nanda, Senior Member IEEE
Dept. of Electronics & Communication Engineering
Malaviya National Institute of Technology Jaipur
Rajasthan-302017, India
Email : sjnanda.ece@mnit.ac.in, +91-9549654237

Biomedical Applications - I

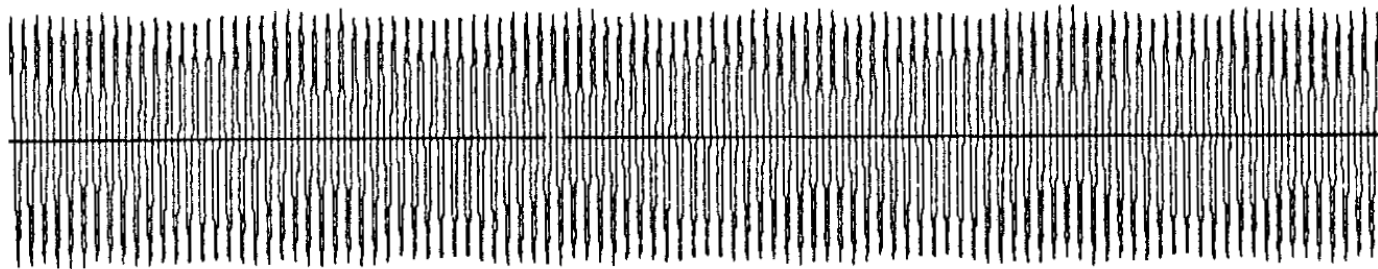


Canceling 50-Hz interference in electrocardiography

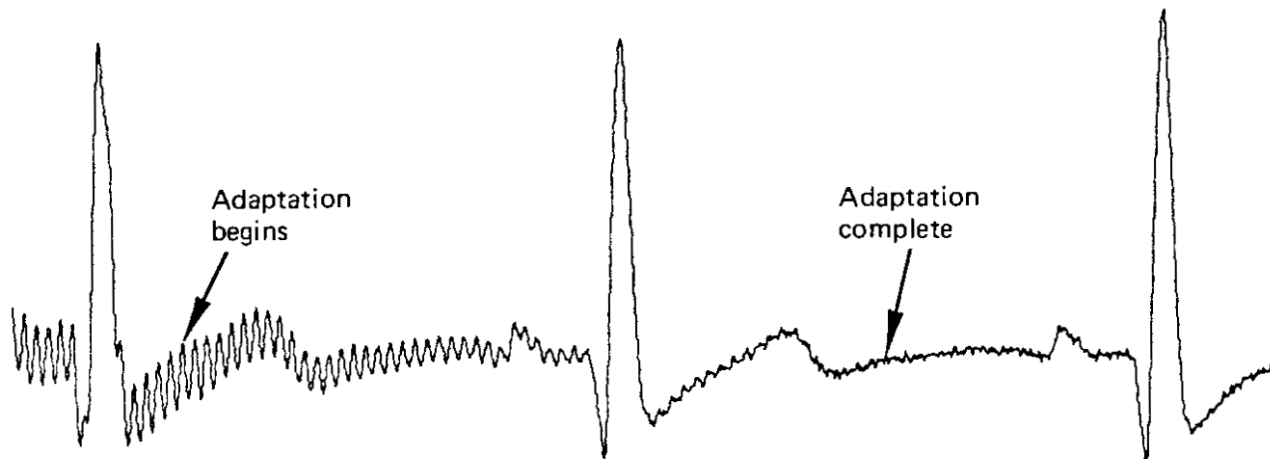
Canceling 50 Hz interference in Electrocardiography



Primary Input
(Obtained
recording)



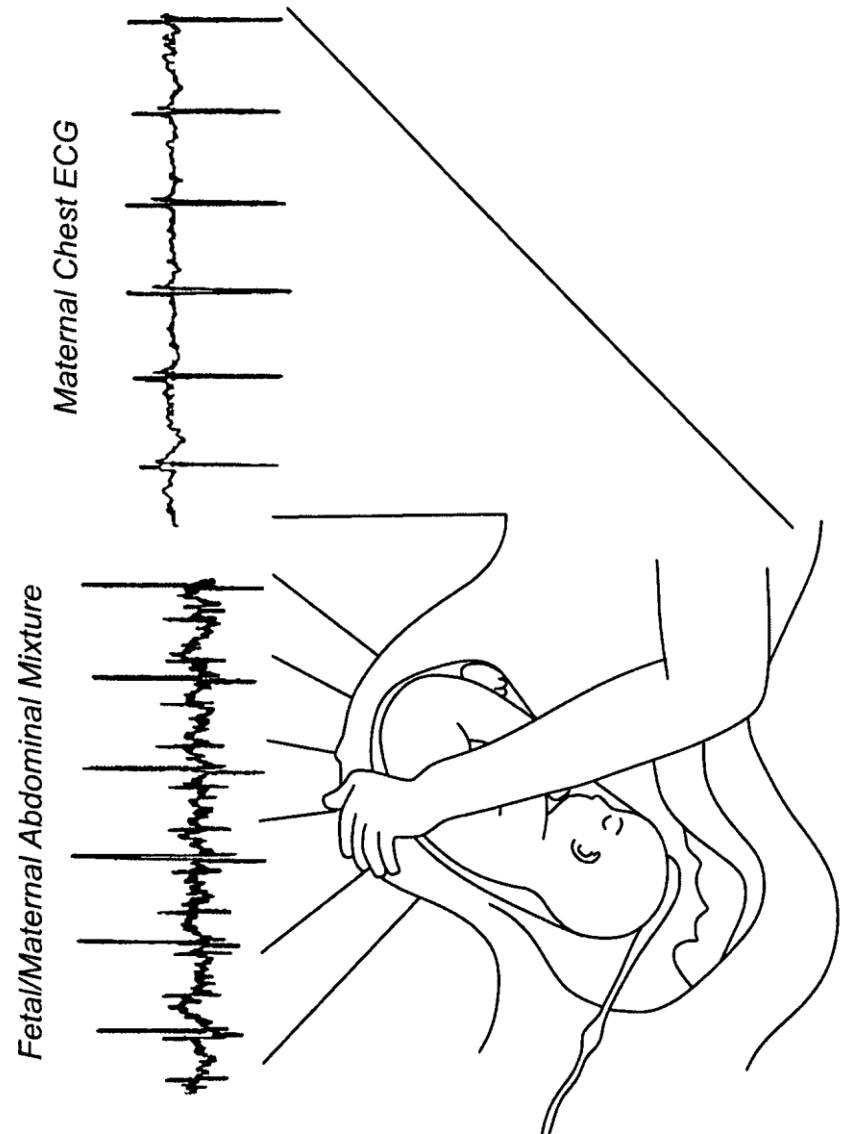
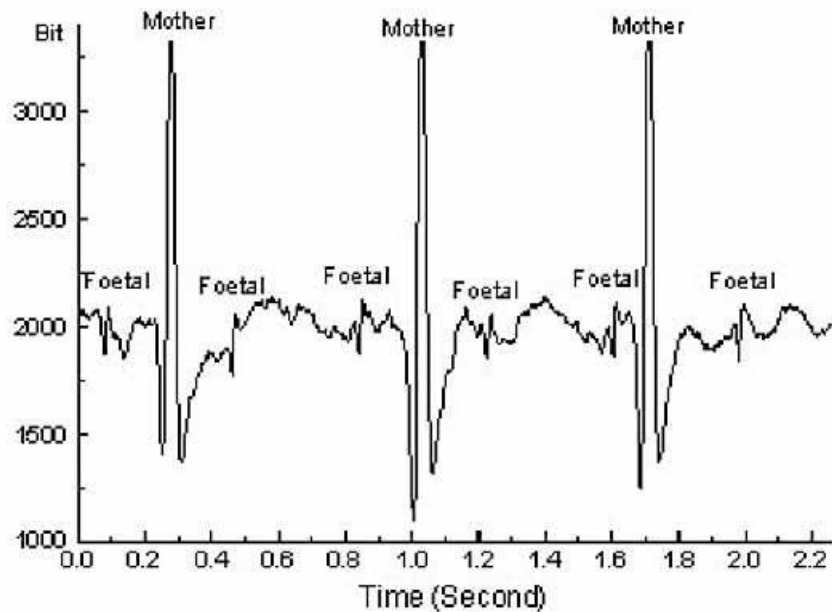
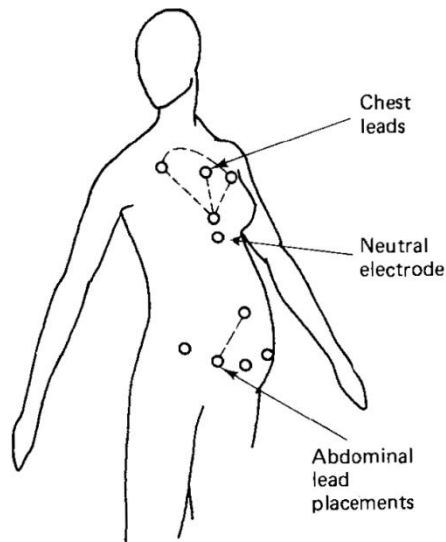
Reference Input
(50 Hz wall outlet)



Output of Neural
Network Filter

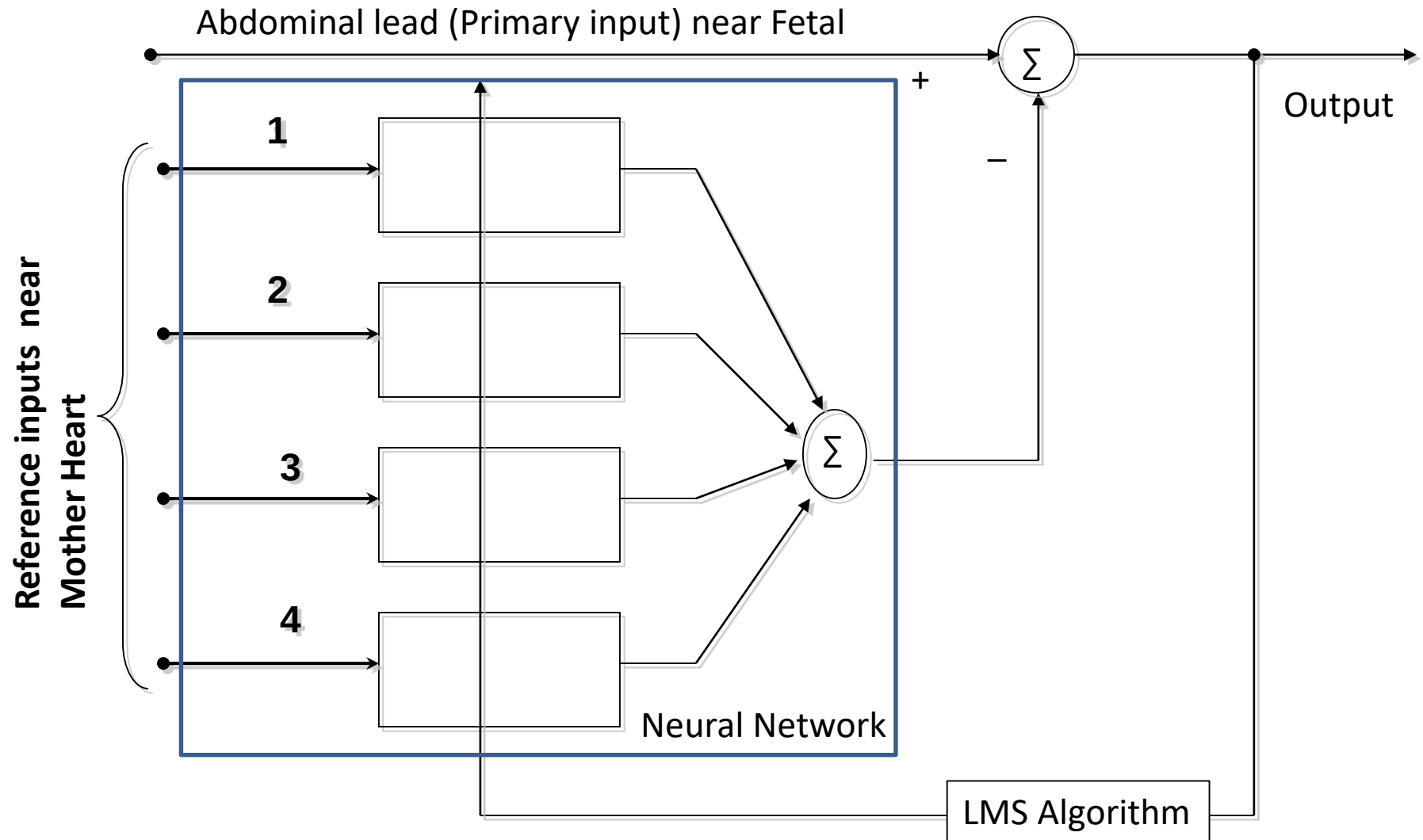
Biomedical Applications - II

Canceling The Maternal ECG In Fetal Electrocardiography

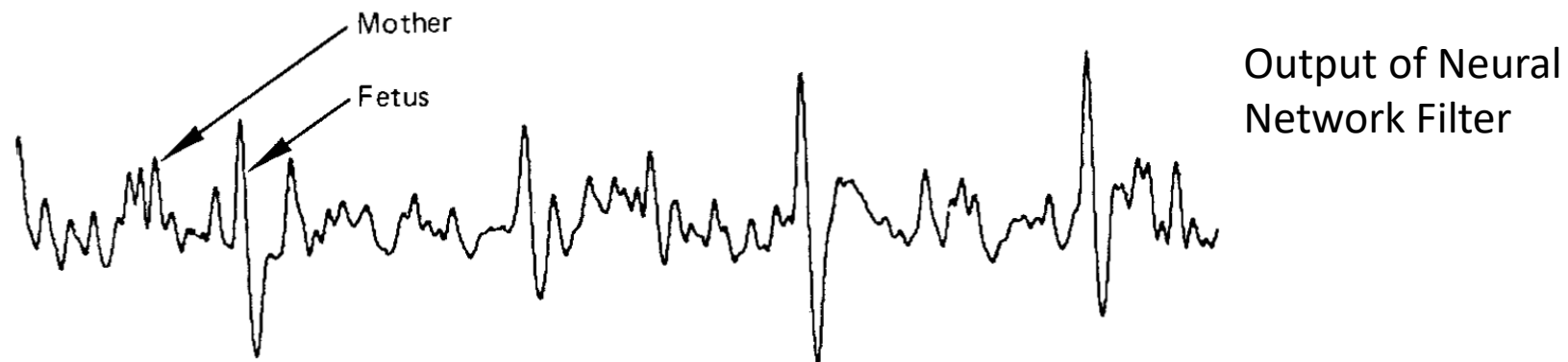
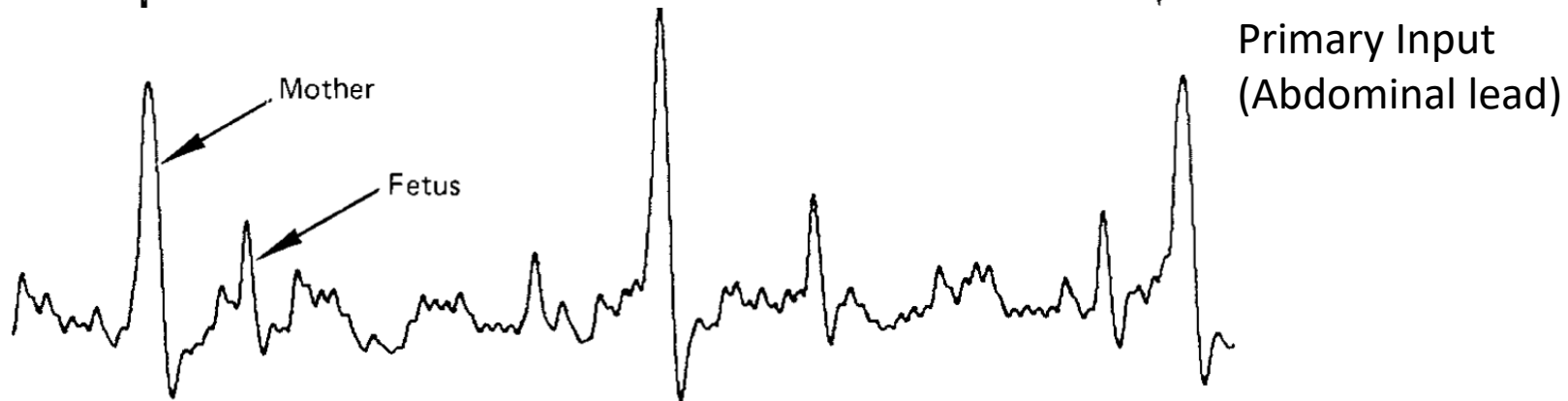
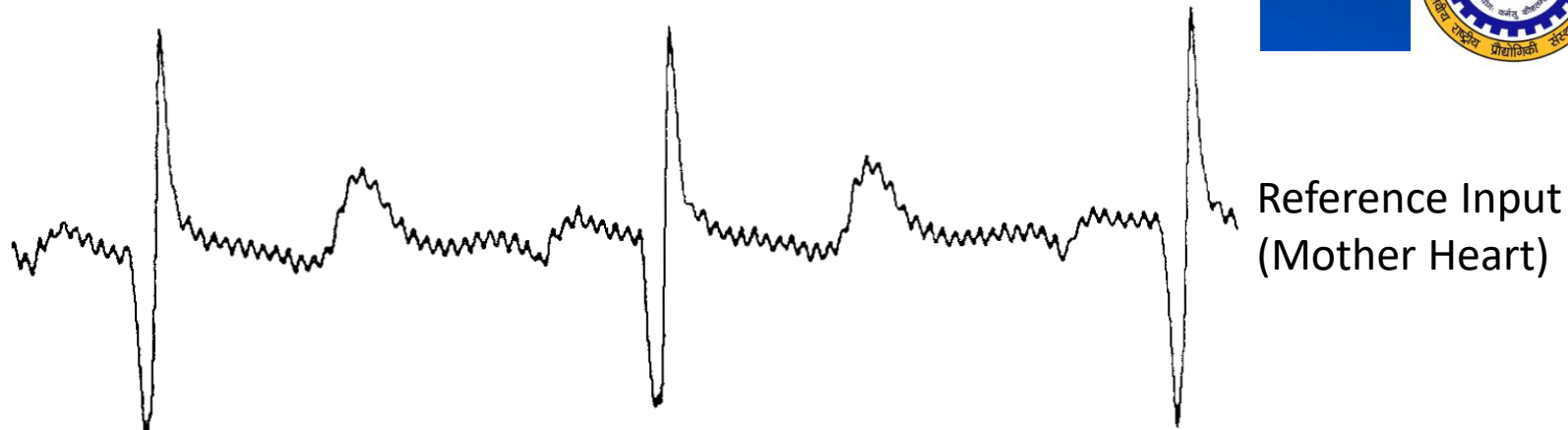


Biomedical Applications - II

Canceling The Maternal ECG In Fetal Electrocardiography

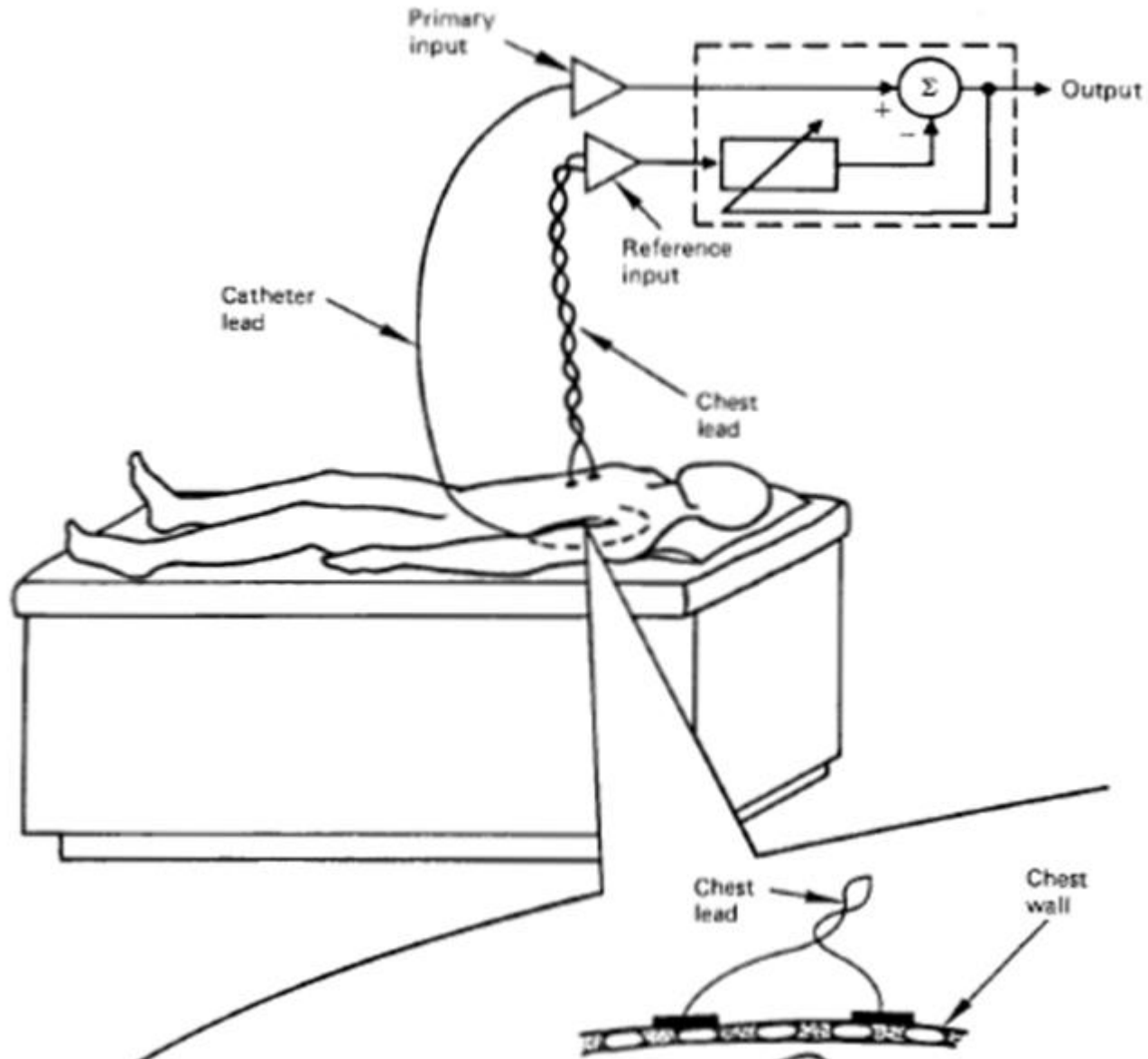


Canceling The Maternal ECG In Fetal Electrocardiography

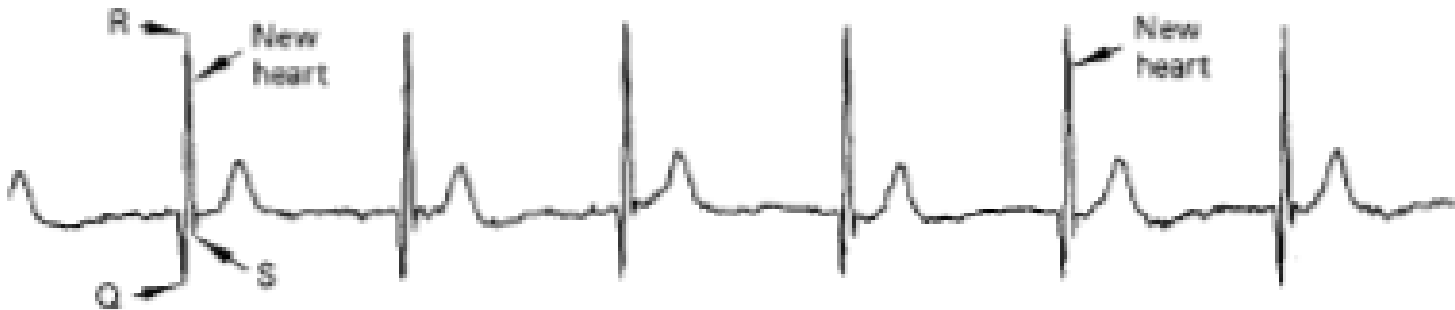


Biomedical Applications - III

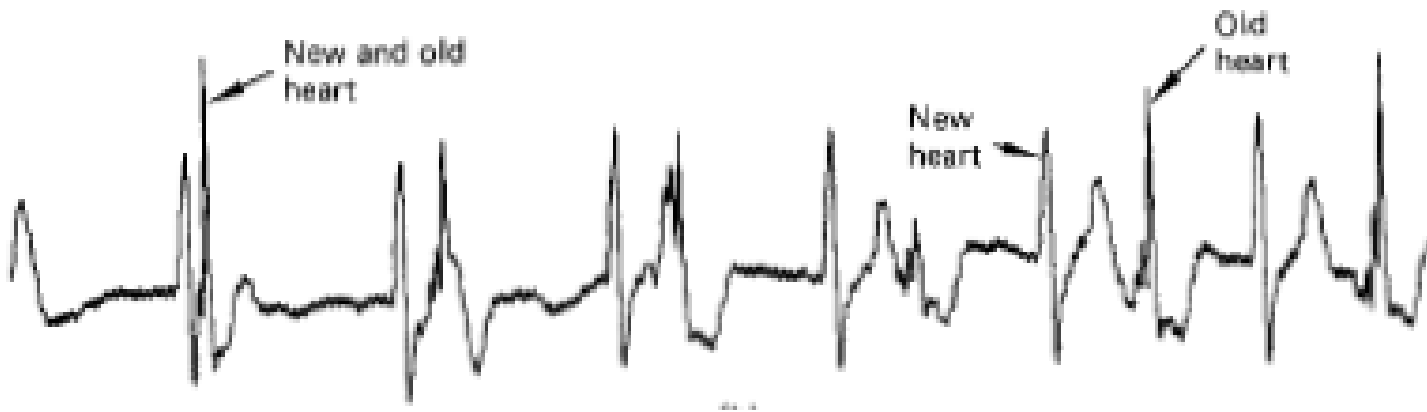
Canceling Donor Heart Interference in Heart transplant Electrocadiography



Canceling Donor Heart Interference in Heart transplant Electrocardiography



Reference Input
(New Heart)

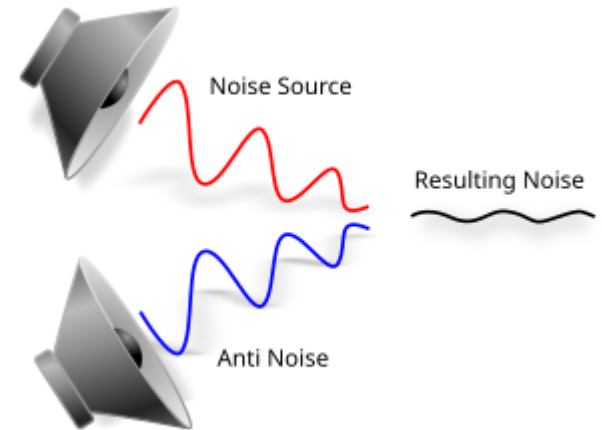
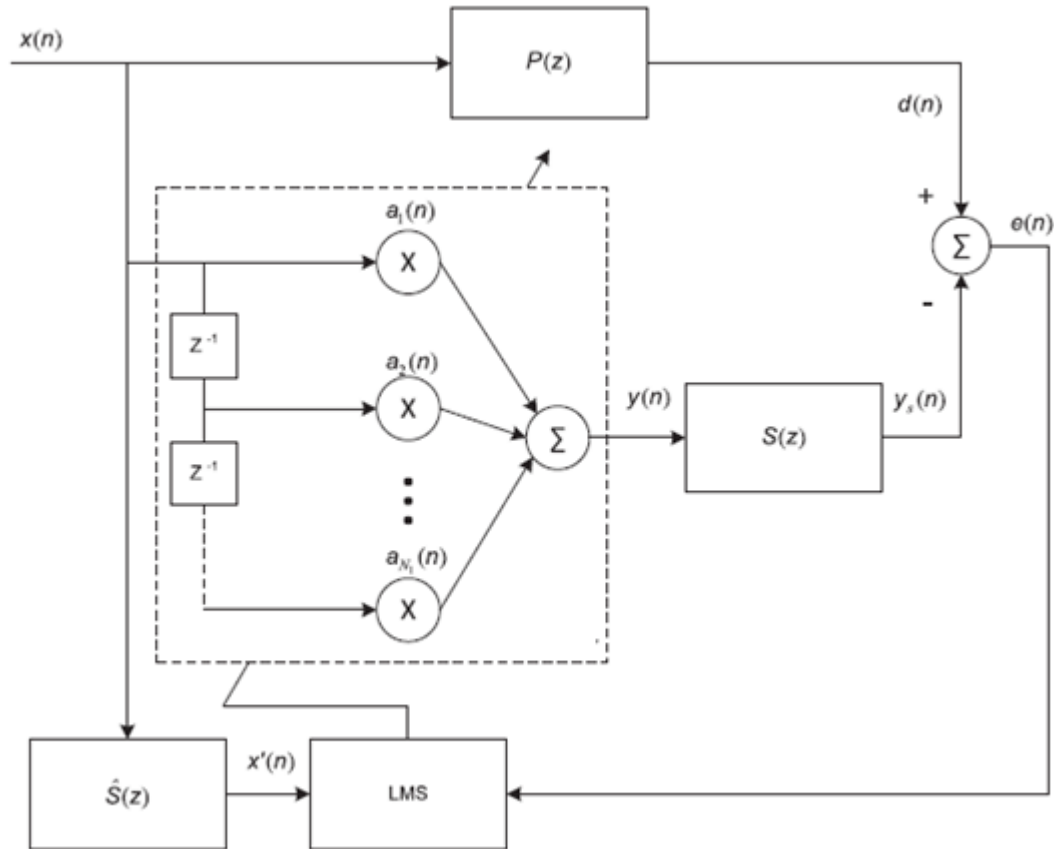
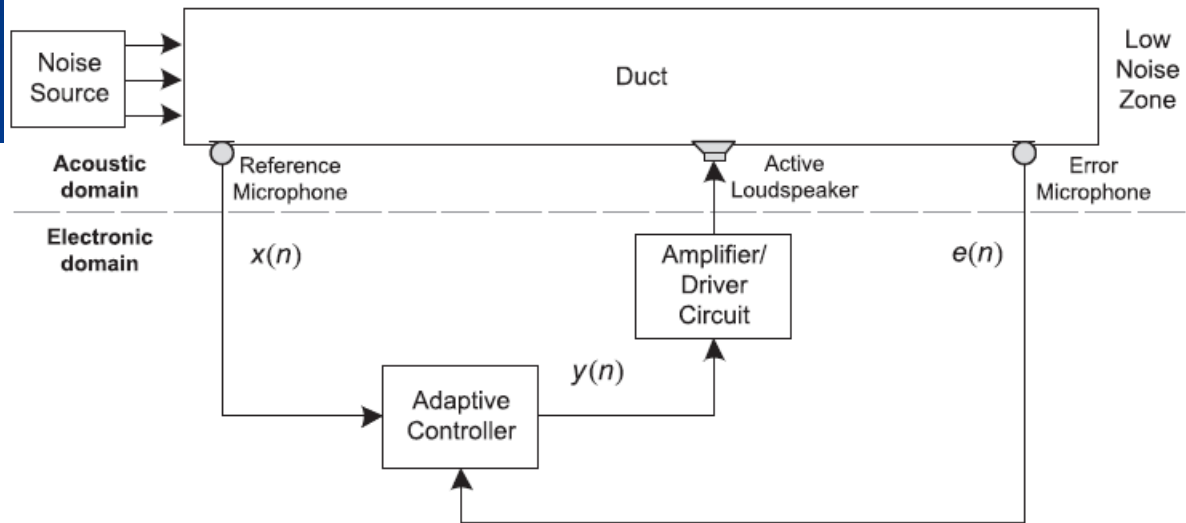


Primary Input
(New + Old Heart)



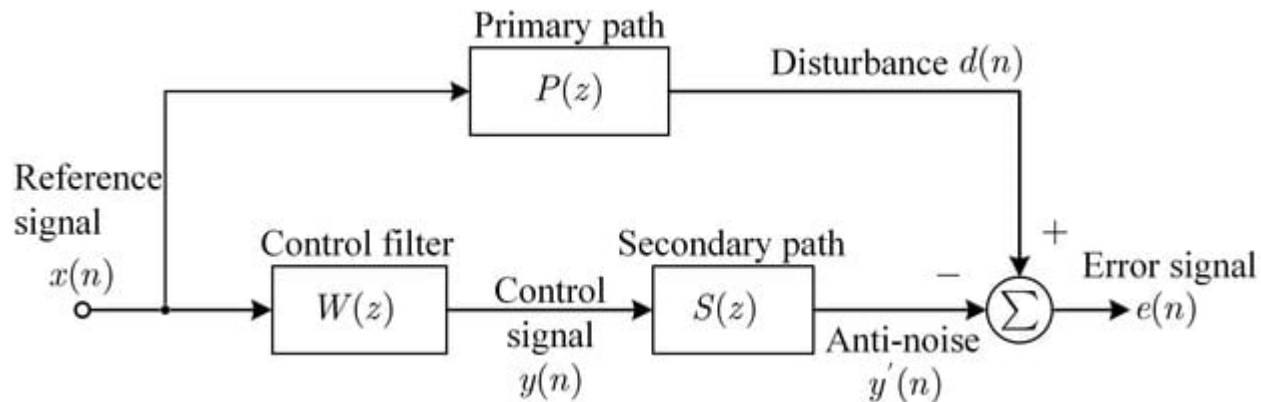
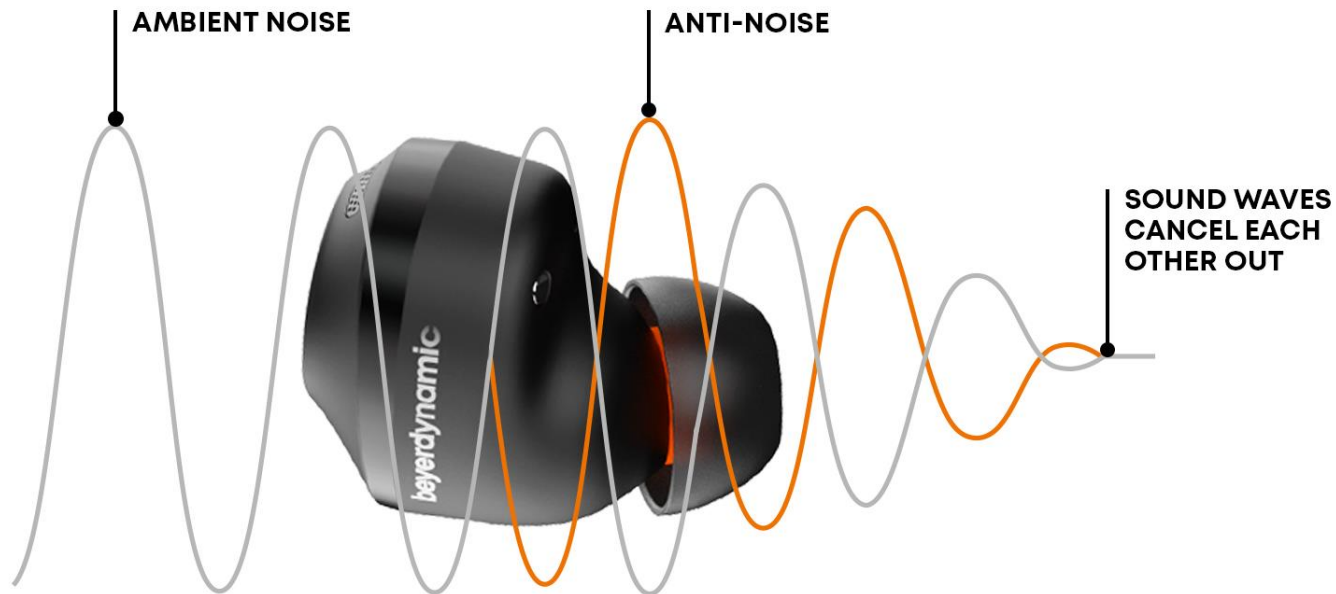
Output of Neural
Network Filter

Active Noise Cancellation

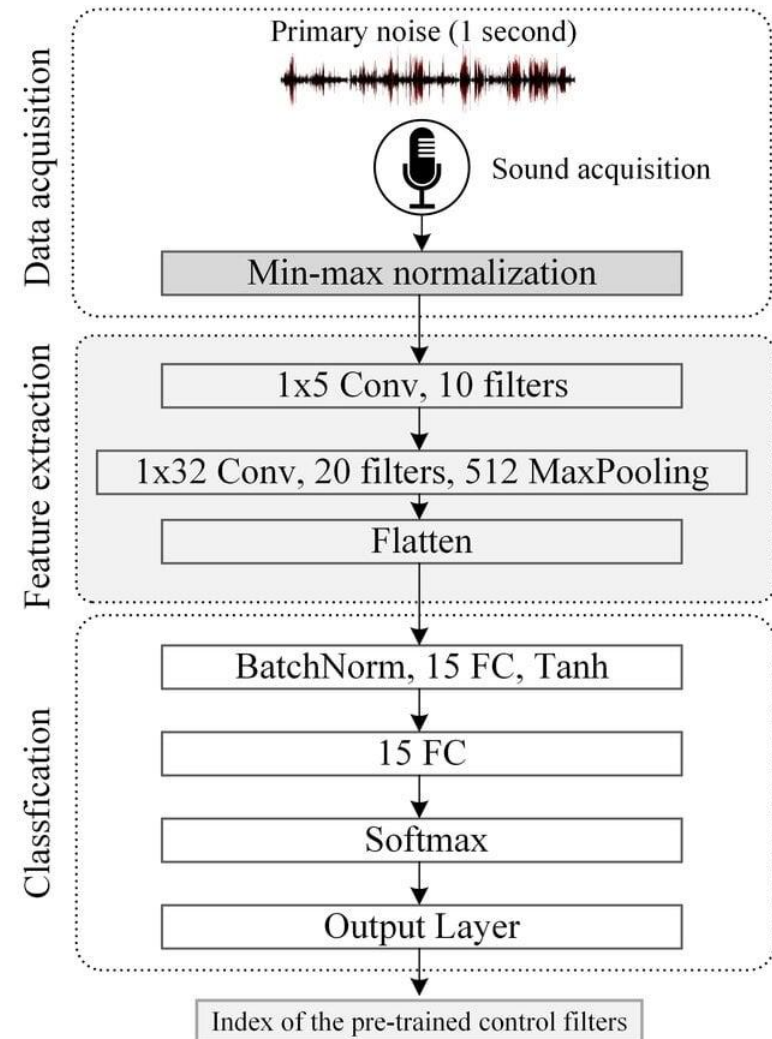
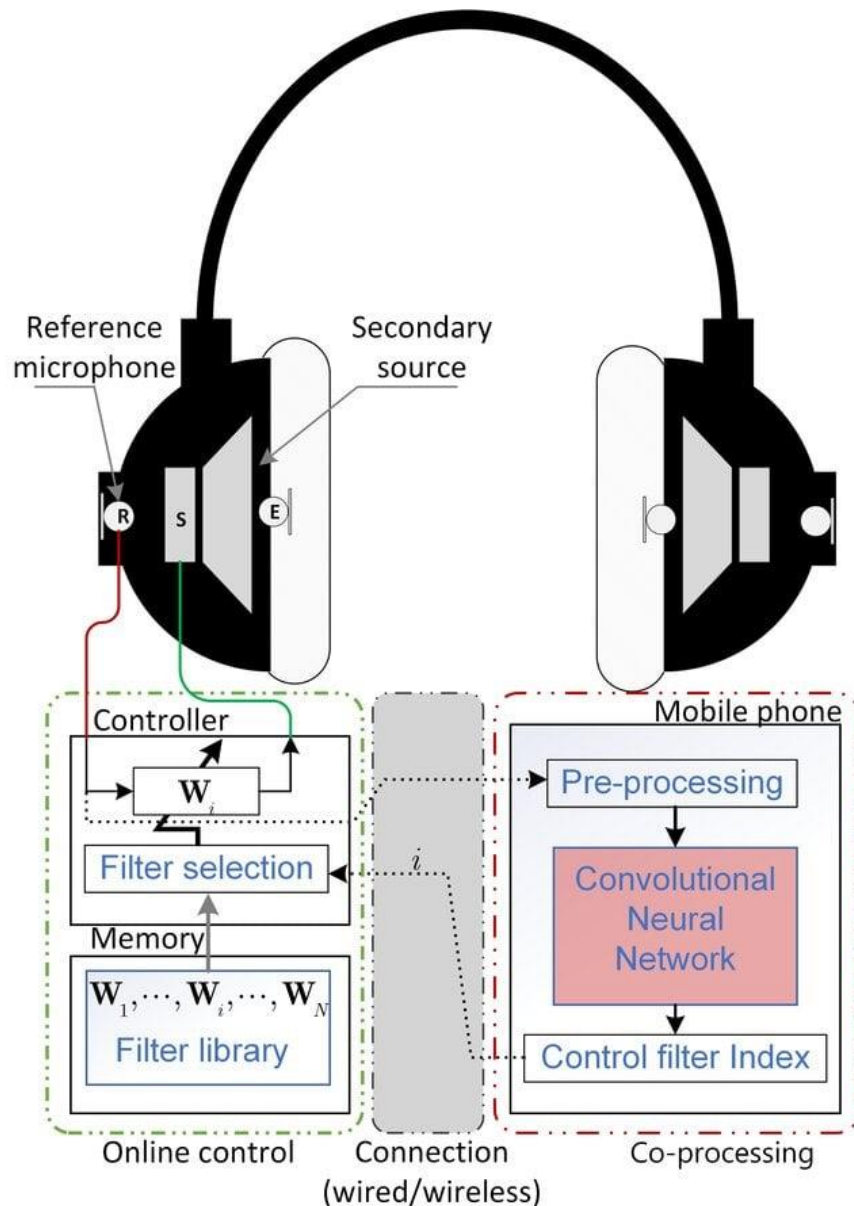


Active Noise Cancellation Headphones

Instead of whining aeroplane engines, humming air-conditioners or disruptive traffic noise, you get perfect tranquillity or soothing music.



Active Noise Cancellation Headphones



Active Noise Cancellation Headphones



Varying noise cancellation based on FxLMS

